Warre Snaet

Physical AI Engineer

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PROFESSIONAL SUMMARY

Joining Captic as a Physical AI Engineer in November 2026. AI Engineering graduate with distinction who builds dependable Python backends, agent infrastructure, applied AI systems, and data pipelines. Experienced with FastAPI, PostgreSQL, computer vision, OCR, RAG, MCP, Azure ML, Docker, Kubernetes, GitHub Actions, and model evaluation.

TECHNICAL EXPERIENCE

Physical AI Engineer | Captic | Ghent, Belgium

November 2026 - present

- Captic, a startup in Ghent, makes AI vision kits for factory robots and machines. Its software finds the next bag or box on a pallet and sends the robot a pick pose, or checks bags for damage. It runs on the machine, not in the cloud.

AI & Software Engineering Intern | 2Ai IPCA | Barcelos, Portugal

February 2026 - June 2026

- Built a Python pipeline that reads echocardiographic DICOM studies, detects burned-in measurement regions, extracts values with OCR and image processing, and exports structured JSON.
- Designed resumable batch processing, validation queues, review tooling, ECG trace extraction, and cross-platform handoff documentation.
- Built a source-video matcher that searches same-exam videos for the frame each measurement came from, and ported OpenCV's ECC alignment to batched PyTorch on CUDA. With faster frame preparation, matching ran 14.9 times faster (1182 s to 80 s on a 39-image benchmark).

Software Engineer, Paid Client Project | Apolloon 24-Hour Run System | Belgium

October 2025 - 2026

- As the sole developer, built a local-first race operations platform with React, TypeScript, Express, Electron, SQLite, and tRPC.
- Replicated the race across three laptops with Raft: majority-vote leader election, automatic failover, UDP discovery on the LAN, and browser screens that move to a live laptop. Tested it with a Raft simulation and failure drills on the event hardware.
- Source code is client-confidential and cannot be shared publicly.

SELECTED PROJECTS

Semi-Supervised Plant Disease Detection in Rust

github.com/SnaetWarre/Research_Project_Rust_Semi-Supervised_Learning

- Built a Burn and Rust pseudo-labeling pipeline using 20% initially labeled data; the saved SSL checkpoint reached 94.90% top-1 accuracy (86.06% supervised baseline) and 94.74% macro F1 on 8,786 held-out images.
- Packaged approximately 916 KB model weights into an approximately 26 MB offline release binary; inference takes about 80 ms per image on an iPhone 12.

Financial AI Agent

github.com/SnaetWarre/Financial_AI_Agent

- In a two-person academic project, owned backend architecture, orchestration, MCP integration, PostgreSQL persistence, Docker delivery, tests, and most frontend integration.
- Built a six-container application with FastAPI streaming, RAG on ChromaDB, eight typed MCP tools, React, local Ollama inference, and explicit trade confirmation.

Azure ML Lifecycle

github.com/SnaetWarre/mlops-sports-ball-classification

- Implemented an Azure ML pipeline with parallel data preparation, cloud training, and model registration, plus FastAPI inference deployed with Docker Compose through GitHub Actions, and Kubernetes manifests.

Athas Code Editor | Open-Source Contributor

github.com/athasdev/athas

- Nine merged pull requests across React, TypeScript, Tauri, and Rust: large-file streaming, input hardening, C# support, image diffing, Linux UI, dynamic proxy ports, resource and memory leak fixes, and an LSP race-condition fix.

EDUCATION

Professional Bachelor in Multimedia & Creative Technologies | AI Engineering track

Howest University of Applied Sciences | September 2023 - June 2026 | Graduated with distinction

Relevant coursework: MLOps 18/20, Azure ML pipelines, Docker, Kubernetes, CI/CD, computer vision, generative AI, reinforcement learning, backend development, and data engineering.

TECHNICAL SKILLS

Backend and data: Python, FastAPI, SQL, PostgreSQL, REST APIs, testing, pandas, NumPy

AI: Computer vision, OpenCV, OCR, RAG, MCP, model evaluation, PyTorch, Burn

Cloud and platform: Azure ML, Docker, Kubernetes, GitHub Actions, Linux, Git

Additional: C#/.NET, TypeScript, React, Rust, Tauri

ADDITIONAL INFORMATION

Languages: Dutch (native), English (C1), French (B1-B2)

Other: Driving licence B; hospitality work since 2021 in fast-paced customer-facing teams.